



Introduction to Bioengineering Theory

(BBL1020; 3-0-0)

Lecture 3

Today's topic

Level of Complexity (reductionism)

Systems Biology

Translational research

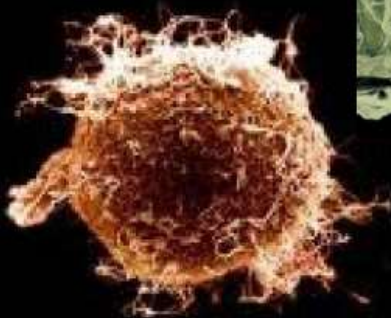
Understanding the level of complexity



**Tissue/
functional subunit**
(10^2 - 10^3 cells)



Cell
(single cell)



Organ
(10^9 - 10^{11} cells)



Whole body (10^{14} cells)



Low

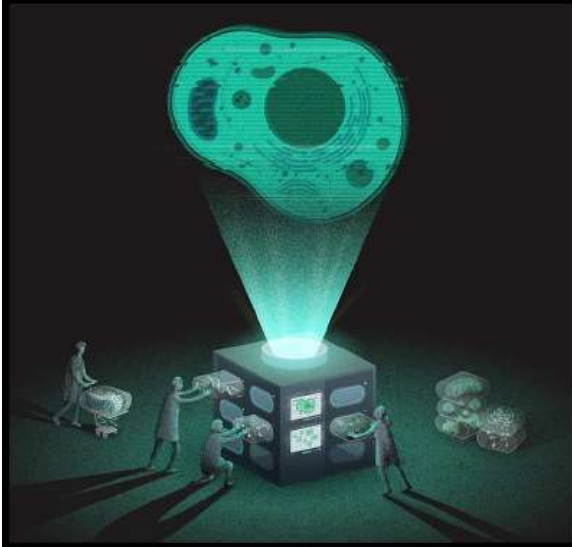
High

Complexity

Understanding the level of complexity

Reductionism

The property of a system is the sum of the properties of its constituents and their local interactions



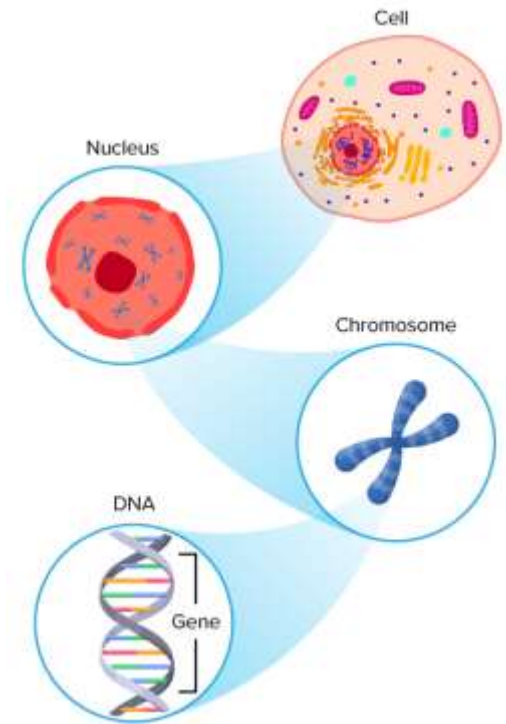
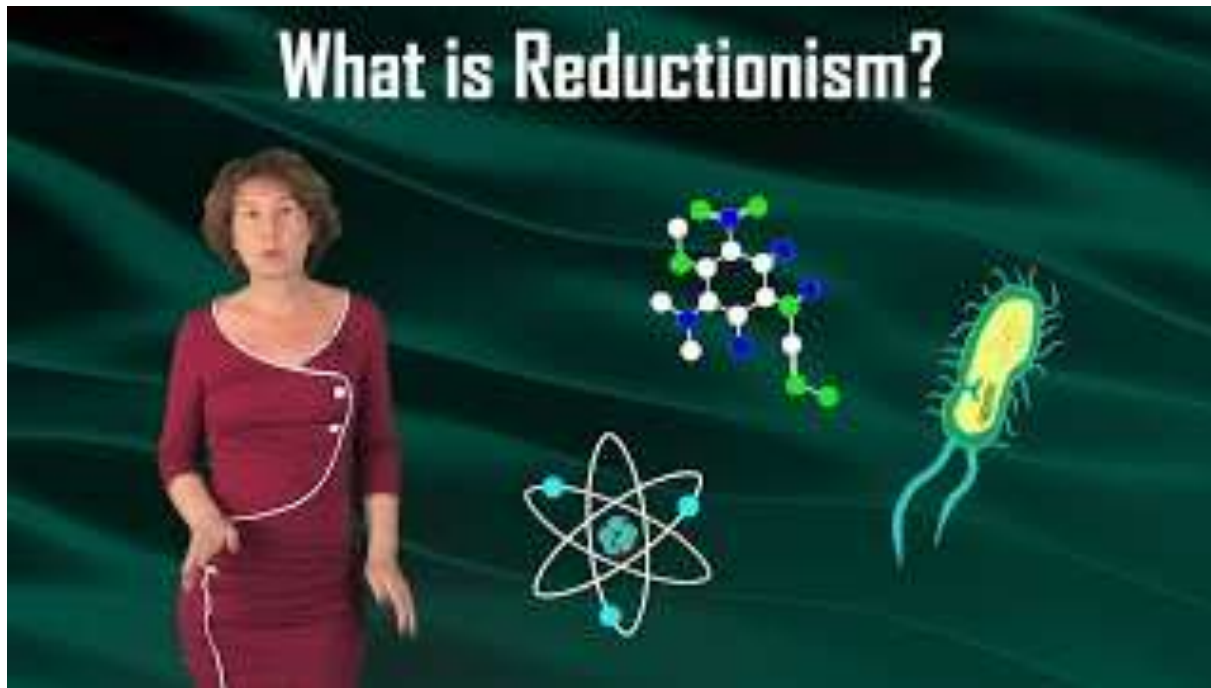
Holism

The property of a system is more than the sum of the properties of its constituents and emerge from the global interactions among themselves under specific context/environment



Reductionism

- Biological reductionism - Reductionism takes upon a mechanistic viewpoint, treating biological systems as analogous to machines with identifiable parts and interactions.



[Understand more](#)

Reductionism

Reductionism

A psychological theory focused on reducing complex phenomena into their most basic parts.



Father of Reductionism - René Descartes (1596–1650)

- Involves analysing systems at different levels of organization to uncover underlying mechanisms that govern their behavior.
- Influence of reductionism extends to interdisciplinary studies that combine various fields to solve complex problems.



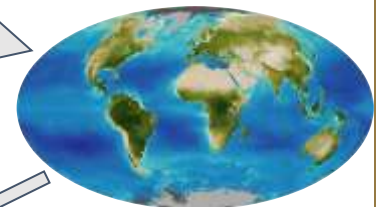
Courtesy: Pablo carlos budassi



Courtesy: Mark Garlick/Science Photo Library/Getty Images



Courtesy: OSSE



Courtesy: NASA

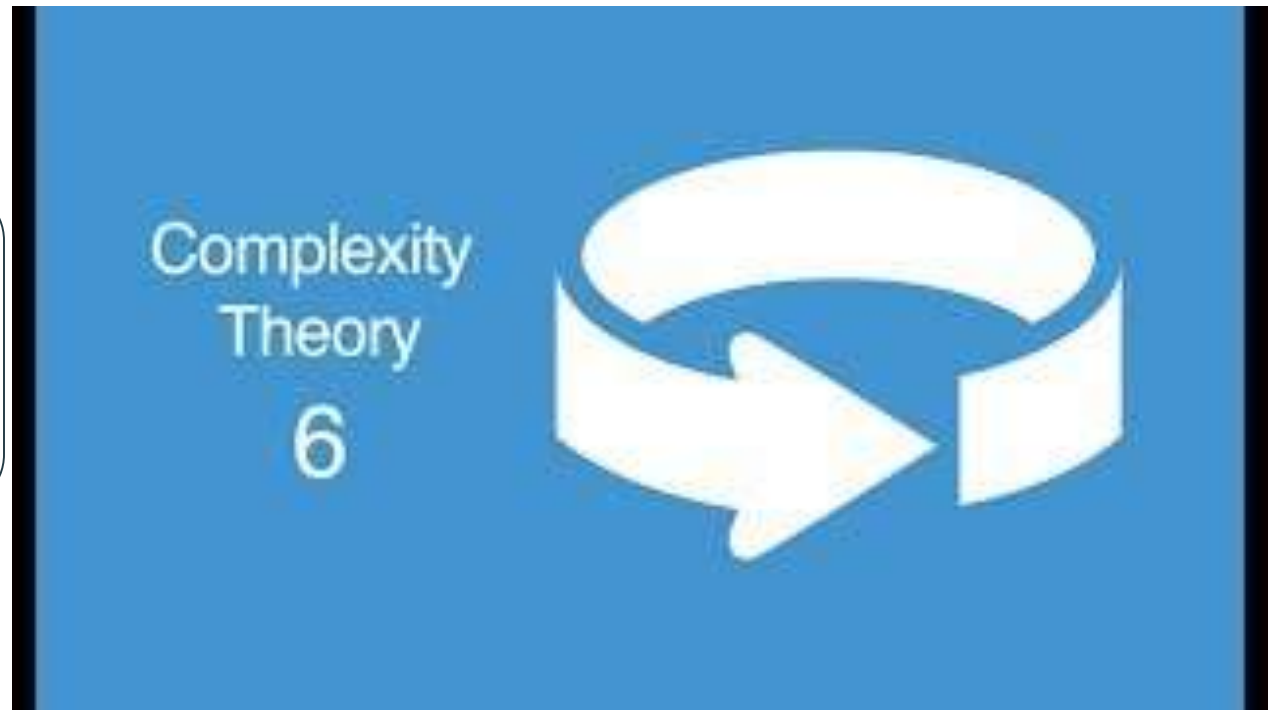


Systems Theory



Systems theory is a conceptual framework based on the principle that the component parts of a system can best be understood in the context of the relationships with each other and with other systems, rather than in isolation.

Systems theory was first introduced in the 1940s by biologist Ludwig von Bertalanffy

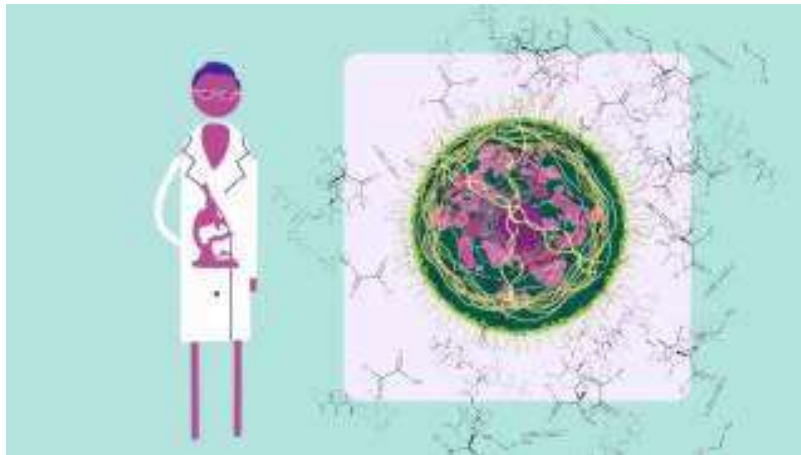


<https://youtu.be/GRnkggRSIDY>

<https://study.com/learn/lesson/systems-theories-in-psychology.html>

Systems Biology

Systems biology aims to understand how biological functions emerge from interactions between the multiple components of living systems by modelling the (dynamics of) interactions and processes.



[watch here](http://www.youtube.com/watch?v=vWSsNi5uFVY)

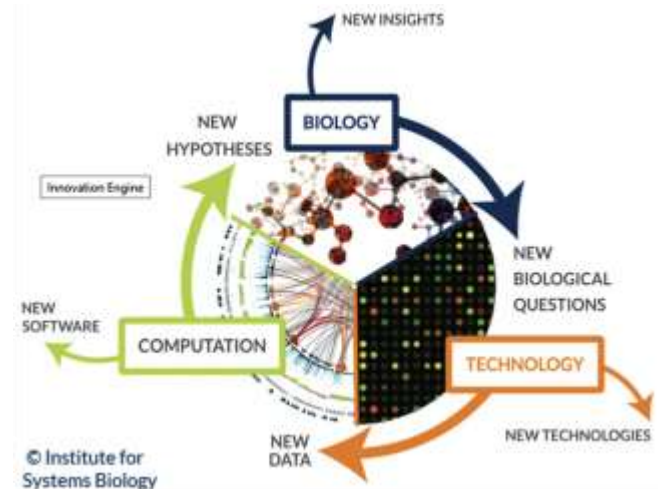
<http://www.youtube.com/watch?v=vWSsNi5uFVY>

<https://elixir-europe.org/communities/systems-biology>

<https://youtu.be/vWSsNi5uFVY>

<https://sci-hub.se/https://doi.org/10.1016/B978-0-12-813762-8.00002-5>

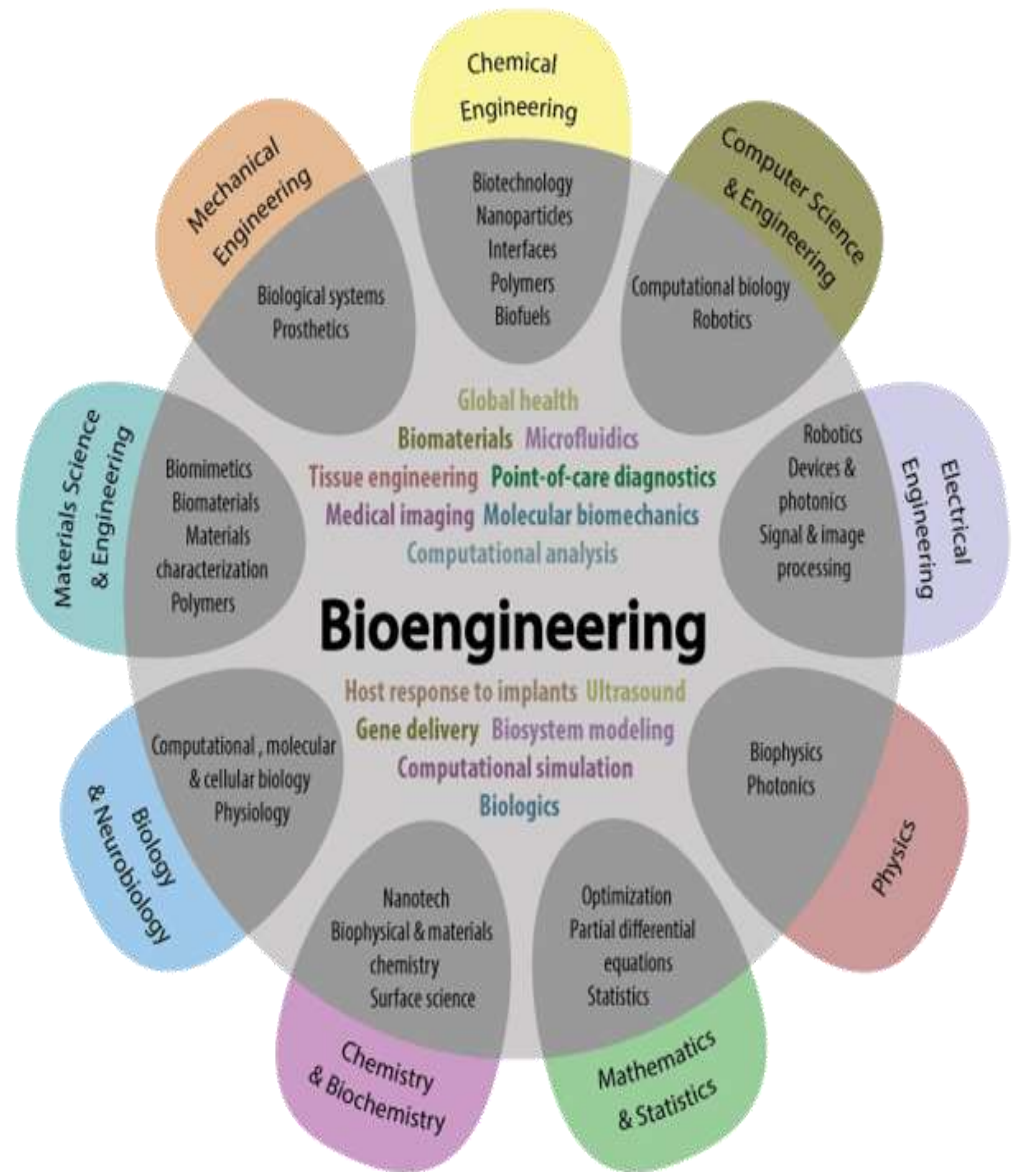
Implementation in 4 steps:



1. Identify and list biomolecules
2. Genome scale maps
3. Conversion of reconstructed maps to models
4. Use them for making Predictive methods

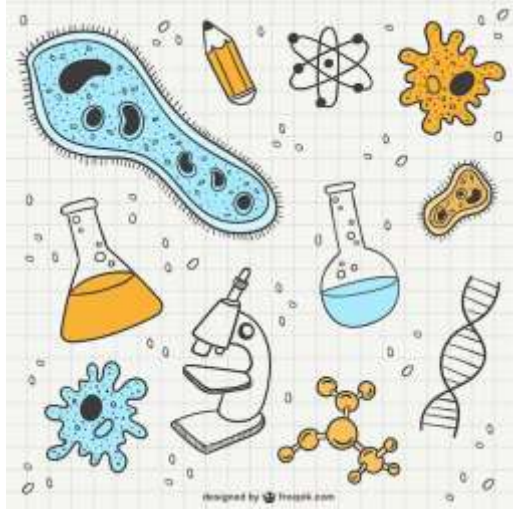
Tinkering with biological systems

Experiment
Learn
Develop
Conceptualize

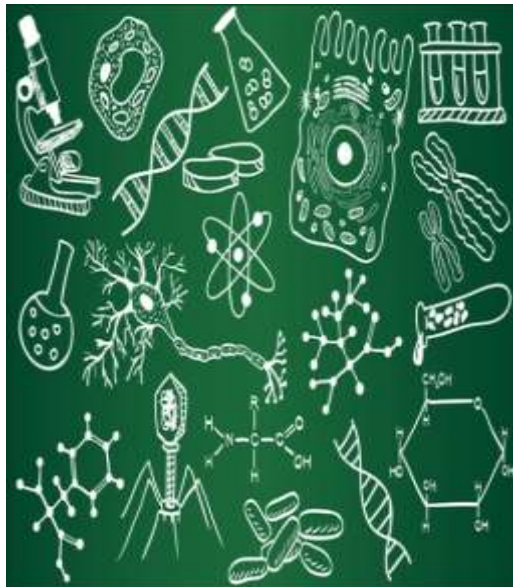


Tinkering with biological systems

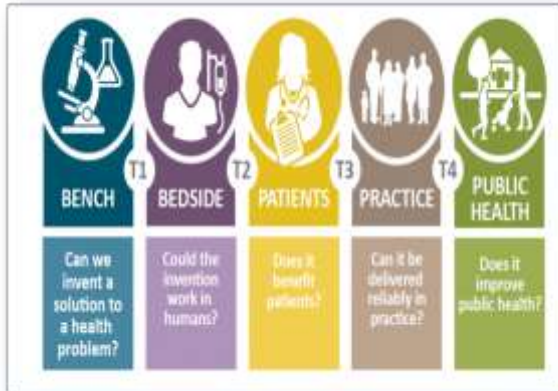
Not just this!



But this !



Translation Research



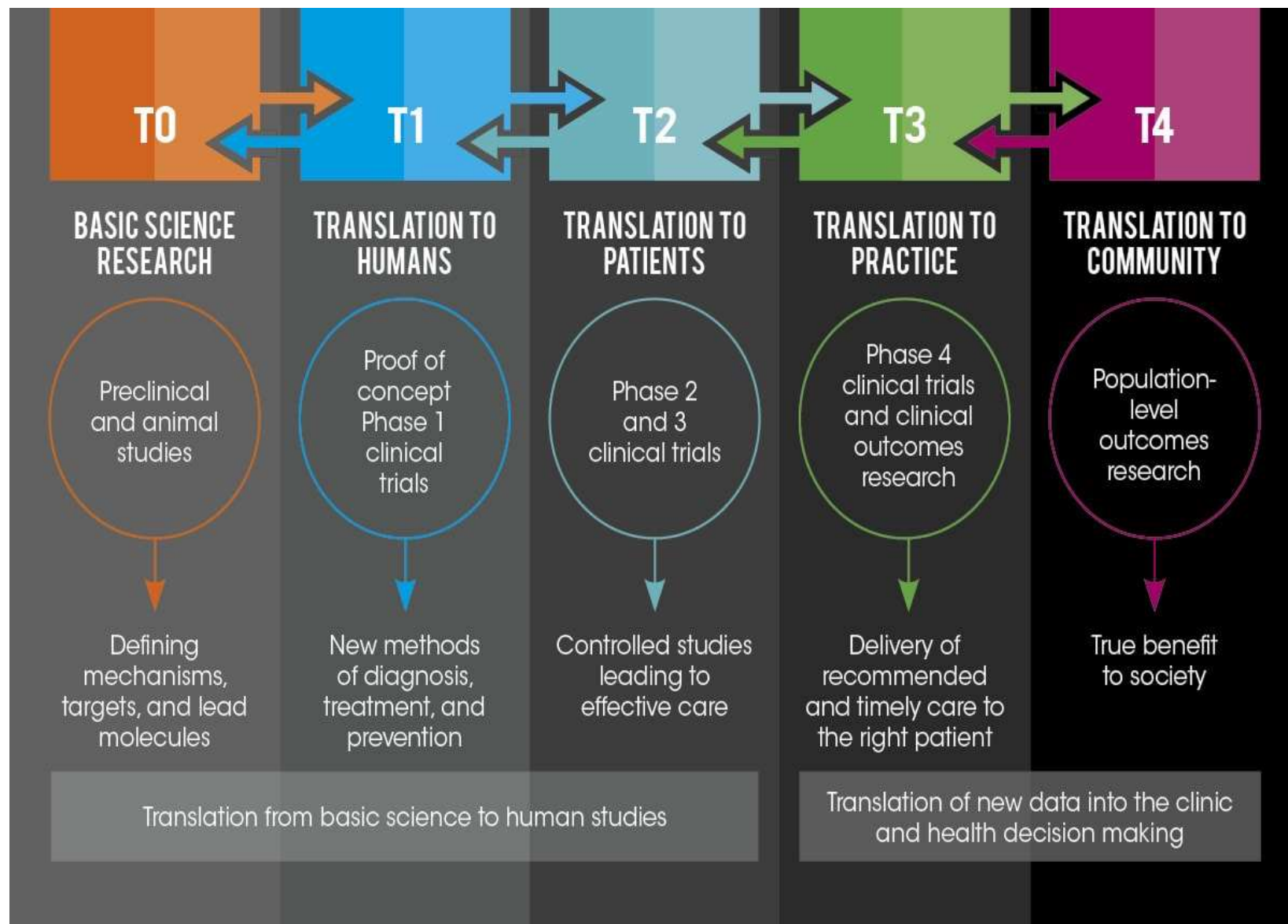
Translational research: refers to the process of translating scientific discoveries from the laboratory into practical applications that benefit human health and society. It involves bridging the gap between basic scientific research and clinical or real-world applications.



[Watch here](http://www.youtube.com/watch?v=TnHLo-hCsgg)

<http://www.youtube.com/watch?v=TnHLo-hCsgg>

Translation Research



Attributes of a bioengineer

A very diverse field and includes all the technologies associated with the application of engineering processes in developing devices, pharmaceuticals, biologics and other products for the prevention and treatment of diseases.

